EE 728: Project Report

Inorganic perovskite ($CsPbX_3$) quantum dot-based light emitting diodes

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Course: Growth and Characterization of Nano electronic Materials
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***Abstract* — We propose an all-inorganic perovskite cesium lead halide ($CsPbX_3$) quantum dot-based LED with a silicon/silicon dioxide hole transporting layer (HTL). We model the optical properties of a quantum dot by solving Schrodinger equations for a 3D quantum well and subsequently working towards derivation of the expected emission wavelength corresponding to electron-hole recombination in the device. The model so developed is applied to the case of cesium lead halide quantum dots and the calculated result is compared against previously observed experimental data published by other groups. Finally, we try to explain the broadening of emission spectrums around peak wavelength on the basis of Lorentzian density of states and depending upon the size of the quantum dot and value of the peak wavelength.**

***Index terms*— QD-LED, Quantum Dot, perovskite, $CsPbX_3$**

I. Introduction

QUANTUM dots (QDs), owing to their unique optical properties like narrow emission spectrum and high luminescent efficiency, have come forward as promising candidates for future lighting and displays. The technology has developed to maturity over the last 25 years and QD-LED display TVs are already in global markets. Much of the study and applications have focused on wurtzite or zinc blende Cd-based QDs.

Perovskites are a broad class of organometallic halides, which in 3D morphology, exhibit an ABX_3 form of crystal structure composed of 5 atoms per unit cell. Recently (since 2016), a lot of research groups have shifted their focus to the use of perovskite QD as the light-emitting layer in LEDs. Traditional organic emitters have a broad emission spectrum (FWHM > 40 nm). Inorganic QD emitters (like CdSe QDs), show relatively narrow FWHM ~ 30 nm but remain size-sensitive and in limited dimension (< 10 nm). Perovskite based LEDs (PeLEDs) are attracting attention from researchers in the field as they ace the emission spectrum with an FWHM < 20 nm. Optical and electronic properties are tunable by varying the composition of constituted halide ions and a smaller degree of the cations. Also, the size of perovskite QDs plays an important role on their optical properties due to quantum confinement effectMost commonly used hybrid organic-inorganic halide perovskites ($CH_3NH_4PbX_3$, i.e. methyl ammonium lead halide) exhibit stability issues. On the other hand, all-inorganic perovskites exhibit higher stability and have huge potential in various optoelectronics.

A common perovskite QD LED consists of an intrinsic active layer embedded in a double-heterojunction structure. An n-type electron transport layer (ETL) and a p-type hole transport layer (HTL) are usually used to construct a typical perovskite QD LED. Under forward bias, the injection of charge carriers into a thin luminescent layer leads to radiative recombination and provides light emission in all directions. Fig. 1, from the work of Kim et. al. [1] reports the energy level diagram of a wide variety of materials being used for HTL, ETL and perovskite.

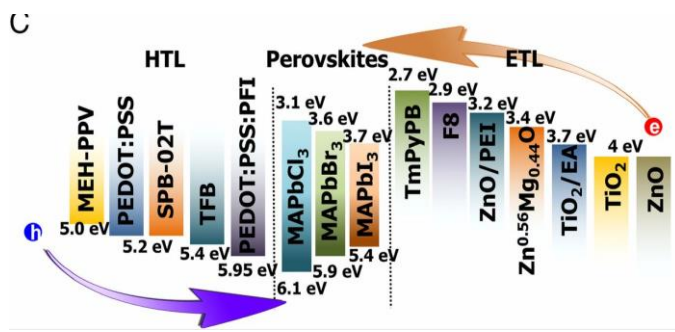

Fig. 1. Energy level diagrams for various ETL, HTL and perovskite materials adapted from [1]

J Song and company [2], for the first time, reported QLEDs based on all-inorganic perovskite cesium lead halide ($CsPbX_3$, X = Cl, Br, I) nanocrystals. The typical QLED stack consisting of ITO/PEDOT:PSS/PVK/QDs/TPBi/LiF/Al was used. Here, we propose a PeLED heterostructure with a $CsPbX_3$ QD active layer and silicon hole transporting layer, inspired by the works of A Gopal [3] and J Song [2]. There are several advantages of using p-type silicon as a hole injection layer, as compared to ITO [4]. ITO is inefficient in injecting holes to polymer layers whereby an additional hole transporting layer is required before an emitting layer in the system. Carrier injection from silicon electrode is improved because most of the electric field drop occurs across the highly resistive layer of thin SiO_2. The SiO_2 on silicon acts as a buffer layer for the system. The buffer layer will restrain the injected hole current and then improve the balance between hole and electron injection. Fig. 2 gives a

schematic of the proposed heterostructure, i.e. Si/SiO_2-/QDs/TPBi/LiF/Al. Average size of the cubic monocrystal QDs synthesized by hot injection method [2] was 8 nm. In the sections ahead, we have developed an analytical model for characterising optical behaviour of QD-LEDs. This model has been applied to our case ($CsPbX_3$) and compared with the experimental results in [2].

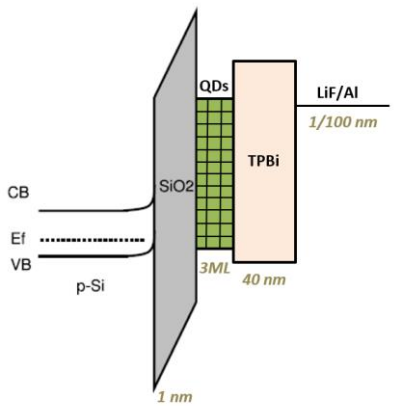

Fig.2. Schematic of the proposed LED heterostructure stack showing relative energy bands

II. Modelling The Optical Properties Of A QD

A quantum dot is ideally a 0-dimensional structure which can be modelled for simplicity as a quantum box. Zero-dimensional structures of other shapes, such as spherical quantum dots, require numerical solution of the Schrodinger equation. A quantum box is a 3D quantum well, i.e. quantum confinement is applied in all the three spatial directions. Consider the box shown in Fig. 3a. For simplicity, let the potential be zero inside the box and infinite elsewhere. The 3D time-harmonic Schrodinger equation for within the box can be given as

$$-\frac{h^2}{8\pi^2 m}\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}\right)\Psi(x,y,z) = E\Psi(x,y,z)$$

The method of separation of variables leads to Eigen functions described by three principal quantum numbers (n_x, n_y, n_z) as follows

$$\Psi(x,y,z) = \sqrt{\frac{8}{L_x L_y L_z}}\, sin\left(\frac{n_x \pi x}{L_x}\right) sin\left(\frac{n_y \pi y}{L_y}\right) sin\left(\frac{n_z \pi z}{L_z}\right)$$

where (n_x, n_y, n_z) take values 1,2,3,…. Eigen energy for a specific Eigen state is given by

$$E_{n_x n_y n_z} = \frac{h^2}{8m}\left(\frac{n_x^2}{L_x^2}+\frac{n_y^2}{L_y^2}+\frac{n_z^2}{L_z^2}\right) \qquad (1)$$

For our case, L_x, L_y, L_z are 8 nm each. Assuming an effective mass of $0.1m_e$ for both electrons and holes based on the work of Mitaya et. al. [5], we can calculate electron energy levels for

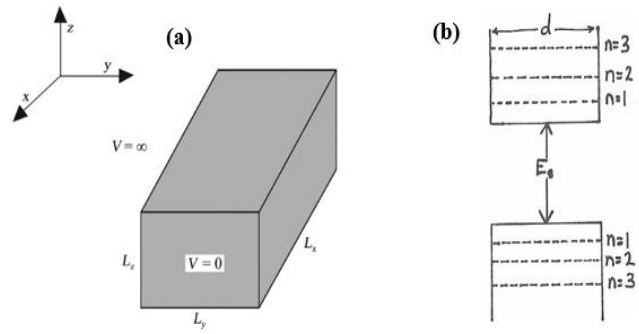

Fig. 3. (a) Schematic of the quantum box (b) Schematic representation of the quantum dot energy bands w.r.t one of the quantum numbers.

different quantum states. Table 1 contains the energy levels for the first 10 quantum states.

Semiconductor nanocrystal quantum dots exhibit unique optical properties due to combination of their material bandgap energy and the quantum confinement effect. When an electron is excited by a photon striking the quantum dot, it behaves as a particle confined in an infinite potential well, since the electron cannot escape from the quantum dot. The hole, created by the excited electron, behaves in the same fashion. Fig. 3b is a schematic diagram of energy states within a quantum dot. The electron, when excited, exists in the conduction band in one of the n energy states, and the hole exists in the valence band, also

n_1	n_2	n_3	Σn_i^2	Degeneracy	Energy
1	1	1	3	None	0.177 eV
1	1	2	6	Threefold	0.354 eV
1	2	1	6		
2	1	1	6		
1	2	2	9	Threefold	0.531 eV
2	1	2	9		
2	2	1	9		
1	1	3	11	Threefold	0.649 eV
1	3	1	11		
3	1	1	11		
2	2	2	12	None	0.708 eV

Table 1. Calculated energy levels for first ten quantum states of the perovskite quantum dots.

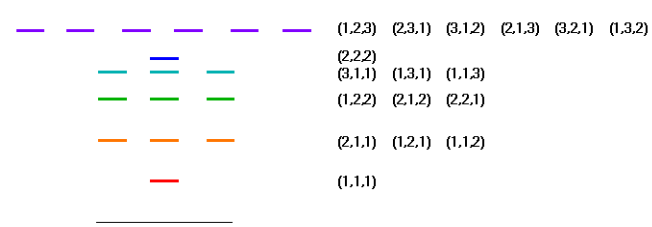

Fig. 4. Energy levels in a 3D Quantum Well and the degeneracy of each quantum state

in one of the n energy states. Therefore, the minimum difference in energy between an electron and hole is not simply the band gap, but the energy difference between n = (1,1,1) energy state of a hole in the valence band, and n = (1,1,1) energy state of an electron in the conduction band. Thus, the expression for energy of an absorbed or emitted photon for a cubic quantum dot of side length d is

$$\frac{hc}{\lambda} = E_{bandgap} + \frac{h^2(n_x^2 + n_y^2 + n_z^2)}{8m_e^* d^2} + \frac{h^2(n_x^2 + n_y^2 + n_z^2)}{8m_h^* d^2} \quad (2)$$

The device we have proposed makes use of the all-inorganic $CsPbX_3$ perovskite quantum dots. S. X. Tao and group in their study [6] have tabulated experimentally observed bandgap values for several perovskites. $CsPbI_3$ showed a bandgap energy $E_{bandgap} = 1.74$ eV. Using this value in equation (2) we can calculate the expected wavelength of emission corresponding to lowest energy (n = (1,1,1)) electron and hole recombination to be 592 nm. This value has been calculated for $d = 8$ nm. Experiments performed by J. Song et. al. [*] on $CsPbX_3$ QD based LED stacks (ITO/PEDOT:PSS/PVK/QDs /TPBi/LiF/Al) with mean quantum dot size of 8 nm showed that for their LED with maximum $CsPbI_3$ concentration, the emission peaked at 584 nm as can also be seen in Fig. 5 taken from their work. This is in close agreement with the value we have obtained from our calculations.

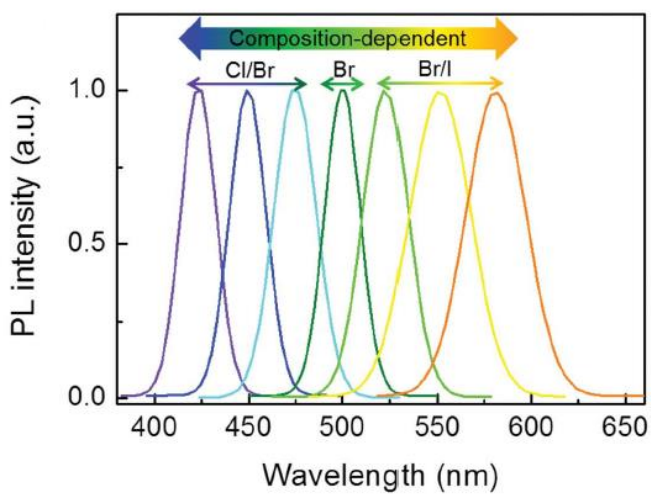

Fig. 5. Spectral plot adapted from [2]. Peak emission wavelength for maximum iodine concentration ($CsPbX_3$) is slightly below 600 nm.

III. Density Of States And Broadening In Emission Spectrum

In this section, we go further with our theoretical analysis of experimental observations. Emission spectrum of an LED is always associated with a broadening around the peak wavelength. A common measure of the broadening is the full width at half the maximum (FWHM). A low value of FWHM implies a sharper emission spectrum and hence, a superior quality of light source. Broadening is observed to be dependent on experimental parameters like temperature and size of the quantum dots being used. Also, the broadening observed at different peak wavelengths is not the same. Emission at wavelengths around the peak wavelength can be attributed to recombination energy being diffused around a central value. Such a broadening around a peak value can be modelled using the Lorentzian density of states given by

$$D(E) = \sum \frac{\frac{\gamma^2}{2\pi}}{\left[(E - E_i)^2 + \left(\frac{\gamma}{2}\right)^2\right]}$$

where, E_i are the discrete energy levels in the quantum device and γ is the broadening coefficient. γ is the measure of the uncertainty in energy of the electron. A reasonable approximation would be, for $\zeta \approx 2 - 5$,

$$\gamma = \frac{\Delta E}{\zeta} \quad (3)$$

Let us now try to get an analytical expression for ΔE. We start with

$$E = \frac{|\boldsymbol{p}|^2}{2m} = \frac{p_x^2 + p_y^2 + p_z^2}{2m}$$

Taking total derivatives on both sides, we get

$$dE = \frac{p_x dp_x + p_y dp_y + p_z dp_z}{m} \quad (4)$$

However,

$$p_x = \frac{hk_x}{2\pi} = \frac{hn_x}{L} \quad (5)$$

where k_x is the x-component of the wave vector corresponding to quantum number n_x, and L is the length of the quantum dot. Similar expressions exist for p_y and p_z. Also, using Heisenberg's uncertainty principle,

$$dp_x \cdot dx \geq \frac{h}{4\pi}$$

$$\therefore dp_x \geq \frac{h}{4\pi L} \quad (6)$$

since L, which is the size of the quantum dot, is the maximum uncertainty in position of the electron. Using values from (5) and (6) in (4) gives

$$dE \geq \frac{h^2}{4\pi m L^2}(n_x + n_y + n_z)$$

and consequently, choosing the minimum possible value of dE in (3) gives us

$$\gamma = \frac{h^2}{4\zeta\pi m L^2}(n_x + n_y + n_z)$$

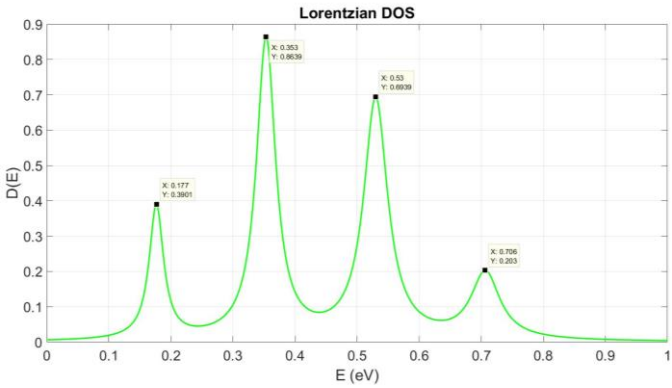

Fig. 6. Plot of Lorentzian density of states for the derived expression of γ. Note that the energy peaks match the values in table 1.

Thus, we observe that the broadening coefficient is a function of size of quantum dot which could be a possible explanation for variation in FWHM with size of quantum dots. Also, it can be seen that broadening corresponding to different quantum states is different. Moreover, while calculating the overall density of states, degeneracy of a particular energy level plays a role. These factors can hint towards difference in FWHM for different emission peaks.

IV. Conclusion

We proposed an all-inorganic perovskite cesium lead halide ($CsPbX_3$) quantum dot-based LED with a silicon/silicon dioxide hole transporting layer (HTL). This design was inspired by the superior stability of all-inorganic perovskite QDs over the hybrid organic-inorganic perovskite QDs, and the possible improvements upon replacing the traditional ITO substrate by a silicon substrate. We modelled the optical properties of a quantum dot by solving Schrodinger equations for a 3D quantum well and subsequently working towards derivation of the expected emission wavelength corresponding to electron-hole recombination in the device. This model was applied to the $CsPbI_3$ quantum dots and wavelength corresponding to minimum energy transition was calculated (592 nm) and found to be close to experimentally observed value (584 nm) in [2]. Finally, we calculated and plotted the Lorentzian density of states for our device using Heisenberg's uncertainty principle and found how the length of the device, quantum state corresponding to transition and degeneracy of transition energy could affect the broadening at a particular peak in the emission spectrum.

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